01–teensy

File: sheets/01–teensy.kicad_sch

02–analog–front

File: sheets/02–analog–front.kicad_sch

03–adc

File: sheets/03–adc.kicad_sch

04–mcp–io

File: sheets/04–mcp–io.kicad_sch

05–cv–dac

File: sheets/05–cv–dac.kicad_sch

06–surface

File: sheets/06–surface.kicad_sch

07–midi

File: sheets/07–midi.kicad_sch

08–connectors–panel

File: sheets/08–connectors–panel.kicad_sch

09–connectors–rear

File: sheets/09–connectors–rear.kicad_sch

10–power

File: sheets/10–power.kicad_sch

11–diagnostics

File: sheets/11–diagnostics.kicad_sch

12–i2c–branches

File: sheets/12–i2c–branches.kicad_sch

Sheet: /
File: pcb–m.kicad_sch

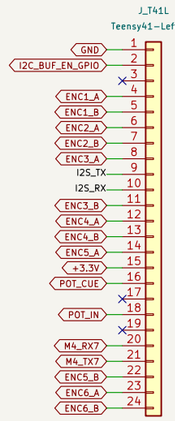
Title:

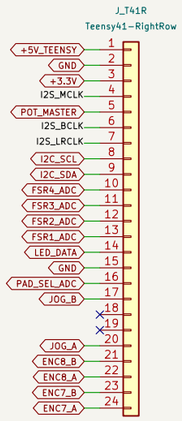
Size: A3	Date:	**Rev:**
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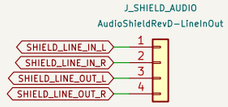
Teensy + Audio Shield: I2S interno al modulo; no microSD ni pot opcional del shield.

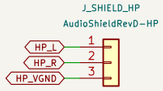
NC: T25/T27/T38/T39 reservados e-drum. HP_VGND nunca se une a GND/AGND.

Filas de sockets conservadas; validar mecanica del montaje antes de fabricar.









Sheet: /01-teensy/
File: 01-teensy.kicad_sch

Title:

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Frente analogico de la portadora Studio conservado (doc25 seccion 6).

D15: selector en panel: escalera 10k/10k a PAD_SEL_ADC. Polo C via J_PAD_SEL (3 hilos).

SW_PAD representa polos A/B en panel, cableados al header 2x4: no montar switch en PCB-M.

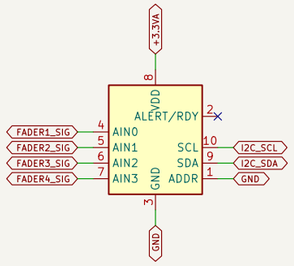
D17: R_ISO_IN_L/R 0 ohm 0805 separate front from codec: remove both if input fails.

R_PAD_L1
33k
LINE_IN_L_F
TAP15_L
R_PAD_L2
12k
TAP15_L
TAP20_L
R_PAD_L3
5.1k
TAP20_L
GND
SW_PADA
PAD 3P3T A/B
PAD_L
LINE_IN_L_F
TAP15_L
TAP20_L
C_AC_L
1uF
PAD_L
AC_L
R_SER_L
1k
AC_L
PREOUT_L
R_BIAS_L
100k
PREOUT_L
PREOUT_L
D_CLAMP_L
BAT54S
AGND
+3.3VA_FRONT
PREOUT_L
JP_BYP_L
BYP L (sin U_OA_IN)
PREOUT_L
BUF_L
U_OA_INA
MCP6004
PREOUT_L
BUF_L
BUF_L
TP_BUF_L
BUF_L
BUF_L
C_IN_L
4.7uF
BUF_L
FRONT_LINE_IN_L
R_PAD_R1
33k
LINE_IN_R_F
TAP15_R
R_PAD_R2
12k
TAP15_R
TAP20_R
R_PAD_R3
5.1k
TAP20_R
GND
SW_PADB
PAD 3P3T A/B
PAD_R
LINE_IN_R_F
TAP15_R
TAP20_R
C_AC_R
1uF
PAD_R
AC_R
R_SER_R
1k
AC_R
PREOUT_R
R_BIAS_R
100k
PREOUT_R
PREOUT_R
D_CLAMP_R
BAT54S
AGND
+3.3VA_FRONT
PREOUT_R
JP_BYP_R
BYP R (sin U_OA_IN)
PREOUT_R
BUF_R
U_OA_INB
MCP6004
PREOUT_R
BUF_R
BUF_R
TP_BUF_R
BUF_R
BUF_R
C_IN_R
4.7uF
BUF_R
FRONT_LINE_IN_R
R_VREF_T
10k
+3.3VA_FRONT
VREF_DIV
R_VREF_B
10k
VREF_DIV
GND
C_VREF
10uF
VREF_DIV
GND
U_OA_INC
MCP6004
VREF_DIV
VREF
VREF
JP_BYP_VREF
BYP VREF (sin U_OA_IN)
VREF_DIV
VREF
TP_VREF
VREF
VREF
U_OA_IND
MCP6004
VREF
OA_IN_D
OA_IN_D
U_OA_INE
MCP6004
+3.3VA_FRONT
V+
GND
C_OA_IN
100nF
+3.3VA_FRONT
GND
C_LOUT_L
10uF
CAP_OUT_L_F
R_LOUT_L
47k
LINE_OUT_L_F
GND
C_LOUT_R
10uF
CAP_OUT_R_F
R_LOUT_R
47k
LINE_OUT_R_F
GND
R_SEL_T
10k
+3.3V
PAD_SEL_ADC
R_SEL_B
10k
PAD_SEL_ADC
GND
R_ISO_IN_L
0
FRONT_LINE_IN_L
CODEC_LINE_IN_L
R_ISO_IN_R
0
FRONT_LINE_IN_R
CODEC_LINE_IN_R

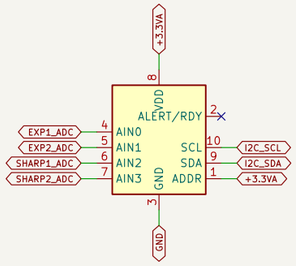
Sheet: /02-analog-front/		
File: 02-analog-front.kicad_sch		
Title:		
Size: A1	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 3/13

ADS1=0x48 (ADDR=GND); ADS2=0x49 (ADDR=VDD); ALERT/RDY no usado.

U_ADS1
ADS1115IDGSR



U_ADS2
ADS1115IDGSR



C_ADS1_VCC
100nF



C_ADS2_VCC
100nF



C_ADS1_BULK
10uF



C_ADS2_BULK
10uF



Sheet: /03-adc/		
File: 03-adc.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 4/13

RESERVADO: MCP23017 0x23 vive en PCB-K. Hoja vacia conservada en la raiz.

Sheet: /04-mcp-io/	
File: 04-mcp-io.kicad_sch	
Title:	
Size: A3	Date:
KiCad E.D.A. 10.0.6	**Rev:** Id: 5/13

CV: 1 + 10k/20k = 1.5; A0 DAC1=GND (0x60), DAC2=3V3 (0x61).

D11: realimentacion antes de 1k serie: BAT54S en nodo protegido hacia jack, a GND/+5V_LED.

DFT: SJ_CV + RC 10 ohm / 47uF -> +5V_CV; swing 4.95V ideal, medir margen/carga.

U_DAC1
MCP4725A0T-E/CH
+3.3V
I2C_SCL
I2C_SDA
GND
SCL
SDA
A0
VDD
VSS
VOUT
DAC1_RAW

U_OA_CVA
MCP6004
DAC1_RAW
CV1_FB
CV1_OUT

R_CV_RF1
10k
CV1_OUT
CV1_FB

R_CV_RI1
20k
CV1_FB
GND

R_CV_SER1
1k
CV1_OUT
CV1_PROTECTED

D_CV_CLAMP1
BAT54S
GND
+5V_LED
CV1_PROTECTED

U_DAC2
MCP4725A0T-E/CH
+3.3V
I2C_SCL
I2C_SDA
GND
SCL
SDA
A0
VDD
VSS
VOUT
DAC2_RAW

U_OA_CVB
MCP6004
DAC2_RAW
CV2_FB
CV2_OUT

R_CV_RF2
10k
CV2_OUT
CV2_FB

R_CV_RI2
20k
CV2_FB
GND

R_CV_SER2
1k
CV2_OUT
CV2_PROTECTED

D_CV_CLAMP2
BAT54S
GND
+5V_LED
CV2_PROTECTED

U_OA_CVC
MCP6004
GND
CV_UNUSED_3
CV_UNUSED_3

U_OA_CVD
MCP6004
GND
CV_UNUSED_4
CV_UNUSED_4

U_OA_CVE
MCP6004
+5V_CV
V+
V-
GND

R_CV_FILTER
10
+5V_CV_IN
+5V_CV

C_DAC1_VCC
100nF
+3.3V
GND

C_DAC2_VCC
100nF
+3.3V
GND

C_OA_CV_VCC
100nF
+5V_CV
GND

C_CV_FILTER
47uF 10V
+5V_CV
GND

C_DAC1_BULK
10uF
+3.3V
GND

C_DAC2_BULK
10uF
+3.3V
GND

C_OA_CV_BULK
10uF
+5V_CV
GND

Sheet: /05-cv-dac/
File: 05-cv-dac.kicad_sch

Title:

Size: A2 | Date: | **Rev:**

KiCad E.D.A. 10.0.6 | Id: 6/13

Componentes de panel por cable. JOG_CLICK via J_PCB_K hacia MCP 0x23.

D13: polo C comun a J_PAD_SEL:2; extremos a .1/.3; contacto central sin cable (1.65V).

D11: pots con RC 1k/100nF del lado ADC. J_HP unico jack HP; HP_VGND aislado de masa.

J_FSR1
JST-PH_2pin_FSR1
+3.3V
FSR1_ADC

J_FSR2
JST-PH_2pin_FSR2
+3.3V
FSR2_ADC

J_FSR3
JST-PH_2pin_FSR3
+3.3V
FSR3_ADC

J_FSR4
JST-PH_2pin_FSR4
+3.3V
FSR4_ADC

R_FSR1
10k
FSR1_ADC
GND

R_FSR2
10k
FSR2_ADC
GND

R_FSR3
10k
FSR3_ADC
GND

R_FSR4
10k
FSR4_ADC
GND

J_JOG
JOG_CABLE
GND
JOG_A
JOG_B
JOG_CLICK

J_POT_MASTER
POT_MASTER_CABLE
+3.3V
POT_MASTER_WIPER
GND

J_POT_CUE
POT_CUE_CABLE
+3.3V
POT_CUE_WIPER
GND

J_POT_IN
POT_IN_CABLE
+3.3V
POT_IN_WIPER
GND

J_PAD_SEL
PAD_SEL_CABLE
+3.3V
PAD_SEL_ADC
GND

J_HP
HP_CABLE
HP_L
HP_R
HP_VGND

R_POT_MASTER
1k
POT_MASTER_WIPER
POT_MASTER

R_POT_CUE
1k
POT_CUE_WIPER
POT_CUE

R_POT_IN
1k
POT_IN_WIPER
POT_IN

C_POT_MASTER
100nF
POT_MASTER
GND

C_POT_CUE
100nF
POT_CUE
GND

C_POT_IN
100nF
POT_IN
GND

Sheet: /06-surface/
File: 06-surface.kicad_sch
Title:
Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 7/13

MIDI TRS Type A: ring=fuente, tip=sumidero; entrada aislada, sleeve IN NC.

D12 CA-033 (2014): R_MIDI_R=33 ohm >=0.5W; R_MIDI_T=10 ohm >=0.25W; sin 100 ohm extra.

JST-PH hacia jacks de panel: 1=tip, 2=ring, 3=sleeve. Sin deteccion de insercion.

D20: ALL DNP / excluded assembly BOM: USB MIDI on PI, no MIDI panel holes. Teensy 0=I2C EN: 1=NC.

J_MIDI_IN
MIDI_IN_PANEL_CABLE
MIDI_IN_TIP
MIDI_IN_RING
1
2
3

U_OPT_MIDI
6N138
MIDI_OPTO_A
MIDI_IN_TIP
+5V_TEENSY
MIDI_OPTO_BASE
MIDI_RX
GND

R_MIDI_IN
220
MIDI_IN_RING
MIDI_OPTO_A

R_MIDI_RX
4.7k
+3.3V
MIDI_RX

C_MIDI_VCC
100nF
+5V_TEENSY
GND

R_MIDI_BASE
10k
MIDI_OPTO_BASE
GND

D_MIDI_IN
1N4148W
MIDI_OPTO_A
MIDI_IN_TIP

J_MIDI_OUT
MIDI_OUT_PANEL_CABLE
MIDI_OUT_TIP
MIDI_OUT_RING
GND
1
2
3

R_MIDI_T
10
MIDI_TX
MIDI_OUT_TIP

R_MIDI_R
33
+3.3V
MIDI_OUT_RING

Sheet: /07-midi/
File: 07-midi.kicad_sch

Title:

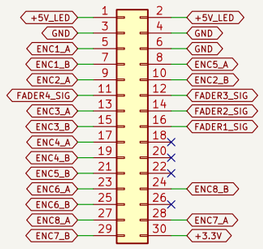
Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 8/13

D8: J1 segun PCB-A fabricada: NC 18(spare)/20(INT_K)/22(INT_A)/26(LED_DIN).

Encoders a Teensy segun firmware; faders al ADS0x48 AIN0-3. LED solo via J_PCB_K.

I2C hacia PCB-A por J2: J1 no lleva SDA/SCL. Retirar pull-ups PCB-A.

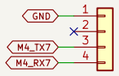
J1_PCBA
IDC_2x15_PCB-A



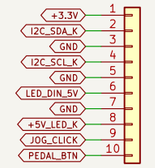
J2_I2C
JST-SH_4pin_I2C



J3_M4
JST-PH_4pin_M4



J_PCB_K
PCB_K_KEY_CABLE



J5_SHARP1
JST-PH_3pin_Sharp1



J6_SHARP2
JST-PH_3pin_Sharp2



J7_EXP1
JST-PH_2pin_EXP1



J8_EXP2
JST-PH_2pin_EXP2



C_SHARP1
10uF



C_SHARP2
10uF



R_EXP1
10k

R_EXP2
10k

Sheet: /08-connectors-panel/
File: 08-connectors-panel.kicad_sch

Title:

Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 9/13

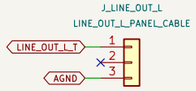
D15: Jacks con tuerca en pared trasera; JST-PH en borde PCB-M, uno por jack.

TRS: 1=tip/2=ring/3=sleeve; TS CV: 1=tip/2=sleeve. HP solo por J_HP (06).

Captura anterior sin contactos de deteccion; no se agrega deteccion de insercion.

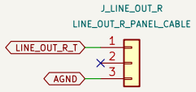
Anillos LINE IN expuestos como redes locales; terminacion balanceada sin definir.



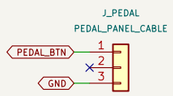


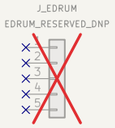












Sheet: /09-connectors-rear/
File: 09-connectors-rear.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

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LDO -> +3.3VA: +3.3V desde Teensy; +5V_LED separado de +5V_TEENSY. D10: NT1 junto a LDO. HP_VGND aislada.

D9: USB-A PI -> J_5V_LED_IN -> F1 0.75A -> +5V_LED. Backup DNP: sin TVS de entrada adecuada en BOM.

D14: pull-ups 2.2k en PCB-M: desoldar R1/R2 de PCB-A. Unica union AGND/GND por NT1.

F1
0.75A MF-MSMF075

U_LDO1
TLV75533PDBV

C_LDO_IN
10uF

C_LDO_OUT
1uF

U_LED_BUF
74AHCT1G125

J_5V_BKUP
5V_BACKUP_DNP

R_SDA
2.2k

R_SCL
2.2k

C_LED_BUF
100nF

J_5V_LED_IN
USB-A_PI_5V_LED_IN

NT1
NetTie_2

#FLG001
PWR_FLAG

#FLG002
PWR_FLAG

#FLG003
PWR_FLAG

#FLG004
PWR_FLAG

#FLG005
PWR_FLAG

#FLG006
PWR_FLAG

Sheet: /10-power/	
File: 10-power.kicad_sch	
Title:	
Size: A2	Date:
KiCad E.D.A. 10.0.6	**Rev:** Id: 11/13

DFT doc24 section 10. Bridges closed by copper: open copper then resolder to restore.

J1_PCBA has no I2C: A branch is J2_I2C. Branch VCC isolation does not prevent signal backfeed.

Select one pull-up pair: 2.2k CLOSED default / 4.7k OPEN. Remove PCB-A R1/R2; PCB-K pull-ups DNP.

TP_CV1/2 before 1k: TP_CV1/2_JACK after 1k. HP_XGND must remain separate.

R_OPT footprints DNP parallel only: MIDI reserved DNP, No change to MIDI OUT CA-033.

SJ_FRONT
SJ_FRONT
+3.3VA
+3.3VA_FRONT

SJ_CV
SJ_CV
+5V_LED
+5V_CV_IN

SJ_LED_K
SJ_LED_K
+5V_LED
+5V_LED_K

SJ_SHARP
SJ_SHARP
+5V_LED
+5V_LED_SHARP

SJ_SDA_A
SJ_SDA_A
I2C_SDA_B_A
I2C_SDA_A_SER

SJ_SCL_A
SJ_SCL_A
I2C_SCL_B_A
I2C_SCL_A_SER

R_SDA_A7
4.7k
PU_SDA_A7
+3.3V

SJ_SDA_K
SJ_SDA_K
I2C_SDA_B_K
I2C_SDA_K_SER

SJ_SCL_K
SJ_SCL_K
I2C_SCL_B_K
I2C_SCL_K_SER

SJ_LED_DATA
SJ_LED_DATA
LED_BUF_OUT
LED_DIN_5V

SJ_PU_SDA_22
SJ_PU_SDA_22
+3.3V
PU_SDA_22

SJ_PU_SDA_A7
SJ_PU_SDA_A7
+3.3V
PU_SDA_A7

R_SCL_A7
4.7k
PU_SCL_A7
+3.3V

R_IND_3V3
1.5k
+3.3V
IND_3V3_A

D_IND_3V3
RED-1mA
GND
IND_3V3_A

R_IND_3V3A
1.5k
+3.3VA
IND_3V3A_A

SJ_PU_SCL_22
SJ_PU_SCL_22
+3.3V
PU_SCL_22

SJ_PU_SCL_A7
SJ_PU_SCL_A7
+3.3V
PU_SCL_A7

R_IND_5V_LED
3.3k
+5V_LED
IND_5V_LED_A

D_IND_3V3A
RED-1mA
AGND
IND_3V3A_A

D_IND_5V_LED
RED-1mA
GND
IND_5V_LED_A

TP_5V_TEENSY
+5V_TEENSY
+5V_TEENSY

TP_5V_RAW
+5V_LED_RAW
+5V_LED_RAW

TP_5V_LED
+5V_LED
+5V_LED

TP_3V3
+3.3V
+3.3V

TP_3V3A
+3.3VA
+3.3VA

TP_FRONT
+3.3VA_FRONT
+3.3VA_FRONT

TP_5V_CV
+5V_CV
+5V_CV

TP_AGND
AGND
AGND

TP_GND
GND
GND

TP_SDA
I2C_SDA
I2C_SDA

TP_SCL
I2C_SCL
I2C_SCL

TP_LED_DIN
LED_DIN_5V
LED_DIN_5V

TP_CV1
CV1_OUT
CV1_OUT

TP_CV2
CV2_OUT
CV2_OUT

TP_CV1_JACK
CV1_PROTECTED
CV1_PROTECTED

R_OPT_BIAS_L
DNP tune parallel
+3.3VA
GND

R_OPT_BIAS_R
DNP tune parallel
+3.3VA
GND

R_OPT_MIDI_RX
DNP tune parallel
+3.3V
MIDI_RX

R_OPT_MIDI_BASE
DNP tune parallel
MIDI_OPTO_BASE
GND

TP_CV2_JACK
CV2_PROTECTED
CV2_PROTECTED

#DFT001
PWR_FLAG
+3.3VA_FRONT

#DFT002
PWR_FLAG
+3.3V

#DFT003
PWR_FLAG
+5V_LED

#DFT004
PWR_FLAG
+5V_LED_RAW

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D21: TCA9517ADGK library graphic/pins = NXP PCA9517ADP C80799 (MSOP8). VCCA=VCCB=3.3V.

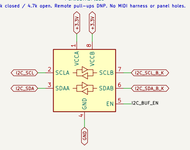
A=local bus shared: B_K and B_A separate cables. NEVER connect two B sides. EN changes only while idle.

EN default pull-up: Teensy pin0 input. Force HIGH: open S1_EN_GPIO BEFORE closing S1_EN_HIGH.

47 ohm 0805 series per cable line; replace with 0 ohm after measuring VOL/rise time.

USBLC6-2SC6 CTS19: 3.5pF max per line: short GND return path. cable connector. PCB-A I2C only via J2.

Pull-ups per segment: 2.2k closed / 4.7k open. Remote pull-ups DNP. No MIDI harness or panel holes.

























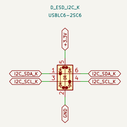




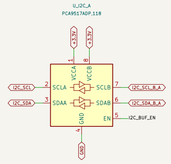




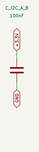












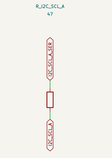










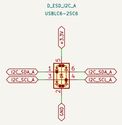




















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